

Predicates and Quantifiers II

Previous Lecture

- *Predicates*
- *Assigning values, universe, truth values*
- *Quantifiers*
- *Quantifiers and negation*
- *Multiple quantifiers*

Quantifiers & predicates in logical formulas

$\forall x \exists y (y \leq x)$

- We can combine predicates, quantifiers, and logical connectives to make more complicated statements

“Every car is blue, or there is a red car”

$P(x)$ - *“car x is blue”*

$Q(x)$ - *“car x is red”*

$$(\forall x P(x)) \vee (\exists x Q(x))$$

“For every number there is a smaller one, or there is the least number”

$x \leq y$ - *“ x is less than or equal to y ”*

$$(\forall x \exists y (y \leq x)) \vee (\exists x \forall y (x \leq y))$$

Quantifiers & predicates in logical formulas (cont.)

- *We can also quantify over compound formulas*

“Every car is red or blue”

$P(x)$ - “car x is red”

$Q(x)$ - “car x is blue”

$$\forall x (P(x) \vee Q(x))$$

“Everyone who has a current password can log onto the network”

$P(x)$ - “ x has a current password”

$Q(x)$ - “ x can logon onto the network”

$$\forall x (P(x) \rightarrow Q(x))$$

Scope of bound variables cont.

● Consider the following, where

$P(x)$ - “car x is red”

$Q(x)$ - “car x is blue”

$\forall x (P(x) \vee Q(x))$

“every car is red or blue”

$\forall x P(x) \vee Q(x)$

“every car is red or car x is blue”

$\forall x P(x) \vee \forall x Q(x)$

“every car is red or every car is blue”

$\forall x (P(x) \vee \forall x Q(x))$

“For every car x , either x is red or all cars are blue”

Example: (logically) defining new predicates

- Can use bi-implication to *logically define* a predicate which is a compound formula over other predicates

$P(x,y)$ - “ y is a parent of x ”

$Q(x)$ - “ x is female”

$S(x,y)$ - “ x and y are sisters”

$$S(x,y) \leftrightarrow Q(x) \wedge Q(y) \wedge \exists z (P(x,z) \wedge P(y,z))$$


Example: Mathematical definitions

Definition (limit):

A number A is a limit of a sequence $\{a_n\}$ if for any number $\varepsilon > 0$ there is N such that for any $n > N$ we have $|a_n - A| < \varepsilon$

$$\lim\{a_n\}, A \leftrightarrow \forall \varepsilon \exists N \forall n ((\varepsilon > 0) \wedge (n > N) \rightarrow (|a_n - A| < \varepsilon))$$

Example: Rules

- *Predicates and quantifiers are (implicitly) present in rules and laws*

“If you have income more than \$20000 you must file a tax report”

$P(x)$ - “ x has income more than \$20000”

$Q(x)$ - “ x must file a tax report”

$$\forall x (P(x) \rightarrow Q(x))$$

Example: Theorems

- *Every theorem involves predicates and quantifiers*

“For every statement there is an equivalent CNF”

$$C(x) - \text{“}x \text{ is a CNF”} \qquad \forall x \exists y (\underline{C(y)} \wedge \underline{(x \Leftrightarrow y)})$$

“A parallelogram is a rectangle if all its angles are equal”

$R(x)$ - *“parallelogram x is a rectangle”*

$A(x)$ - *“all angles of x are equal”*

$$\forall x (A(x) \rightarrow R(x))$$

Examples

● Professor's Carlson class has 29 students of which:

- 3 physics majors are juniors,
- 2 electrical engineering majors are juniors,
- 4 math majors are juniors,
- 12 physics majors are seniors,
- 4 el engineering majors are seniors,
- 2 el engineering majors are grad students,
- 2 math majors are grad students.

p – physics major
m – math major
e – el eng major
j – junior
s – senior
g – grad

Express symbolically and evaluate the truth value:

1) There is a math major in the class who is a junior

$$\exists x (m(x) \wedge j(x))$$

T

Examples

● Professor's Carlson class has 29 students of which:

- 3 physics majors are juniors,
- 2 electrical engineering majors are juniors,
- 4 math majors are juniors,
- 12 physics majors are seniors,
- 4 el engineering majors are seniors,
- 2 el engineering majors are grad students,
- 2 math majors are grad students.

p – physics major
 m – math major
 e – el eng major
 j – junior
 s – senior
 g – grad

Express symbolically and evaluate the truth value:

2) There is a senior in the class who is not a math major

$$\exists x (s(x) \wedge \neg m(x)) \quad T$$

Examples

● Professor's Carlson class has 29 students of which:

- 3 physics majors are juniors,
- 2 electrical engineering majors are juniors,
- 4 math majors are juniors,
- 12 physics majors are seniors,
- 4 el engineering majors are seniors,
- 2 el engineering majors are grad students,
- 2 math majors are grad students.

p – physics major
 m – math major
 e – el eng major
 j – junior
 s – senior
 g – grad

Express symbolically and evaluate the truth value:

3) Every student in the class is majoring in math or physics

$$\forall x (m(x) \vee p(x)) \quad F$$

Examples

● Professor's Carlson class has 29 students of which:

- 3 physics majors are juniors,
- 2 electrical engineering majors are juniors,
- 4 math majors are juniors,
- 12 physics majors are seniors,
- 4 el engineering majors are seniors,
- 2 el engineering majors are grad students,
- 2 math majors are grad students.

p – physics major
 m – math major
 e – el eng major
 j – junior
 s – senior
 g – grad

Express symbolically and evaluate the truth value:

4) No grad student in the class is a physics major

$$\forall x (\neg (g(x) \wedge p(x))) \quad T$$

Examples

● Professor's Carlson class has 29 students of which:

- 3 physics majors are juniors,
- 2 electrical engineering majors are juniors,
- 4 math majors are juniors,
- 12 physics majors are seniors,
- 4 el engineering majors are seniors,
- 2 el engineering majors are grad students,
- 2 math majors are grad students.

p – physics major
 m – math major
 e – el eng major
 j – junior
 s – senior
 g – grad

Express symbolically and evaluate the truth value:

5) Every senior in the class is majoring in either physics or el engineering

$$\forall x (s(x) \rightarrow (p(x) \vee e(x))) \quad T$$

Interpretation

- A logic statement is meaningless

It only makes sense if we specify the domains and particular meanings of predicates

$$\forall x P(x)$$

Interpretation

domain: animals
 $P(x)$: x has horns



domain: cars
 $P(x)$: x is red



domain: numbers
 $P(x)$: x is even



Logical Equivalence of Predicates

- *Recall that two compound propositions Φ and Ψ are logically equivalent ($\Phi \Leftrightarrow \Psi$) if and only if $\Phi \leftrightarrow \Psi$ is a tautology.*
- *In other words, propositional logical equivalence means "equivalent regardless of the truth of propositional variables"*
- *In predicate logic, there are two types of unknowns:*
 - *The predicates themselves*
 - *The indeterminates*
- *How can we define logical equivalence?*

Logical Equivalence of Quantified Propositions

● Definition:

Two formulas in predicate logic are said to be *logically equivalent* if they are equivalent **for any** interpretation of predicates and any domain for their variables.

● E.g. De Morgan's laws for quantifiers:

$$\neg(\forall x P(x)) \iff \exists x \neg P(x)$$

$$\neg(\exists x P(x)) \iff \forall x \neg P(x)$$

Proving logical Equivalence

- *Have to show equivalence for any interpretation of predicates*
- *Do so by assuming an arbitrary predicate (i.e. making no assumptions about its meaning or domain), then **generalizing** it to all predicates*
- *E.g. $\neg(\forall x P(x)) \iff \exists x \neg P(x)$*
 - *Assume P is an arbitrary predicate.*
 - *If $\neg(\forall x P(x))$, then it is not the case that every value a satisfies $P(a)$*
 - *Pick one such a . Now $\neg P(a)$ holds, so $\exists x \neg P(x)$ does too*
 - *If on the other hand, $\exists x \neg P(x)$, then there exists a such that $\neg P(a)$ holds*
 - *So, it is not the case that $P(a)$ is true for all a , i.e. $\neg(\forall x P(x))$ holds*
 - *Therefore, equivalence holds for any interpretation of P*

Implicit assumptions

- *Have to be careful with these types of proofs! You may be making **implicit assumptions** about the interpretation*
- *What's wrong with this argument? Show $\forall x P(x) \rightarrow \exists x P(x) \Leftrightarrow T$*
 - *Assume P is arbitrary*
 - *If $\forall x P(x)$, then for every value a in the domain, $P(a)$ holds*
 - *Pick one value a in the domain such that $P(a)$ holds*
 - *Then $\exists x P(x)$ holds*

Practice

Exercises from the Book:

7th edition: No. 34, 38, 45, 50 (page 55 – 56), No. 25, 26 (page 67)

8th edition: No. 34, 40, 47, 52 (page 58 – 59), No. 25, 26 (page 71)

Represent in symbolic form

a definition “Jaywalk means to cross a roadway, not being a lane, at any place which is not within a crosswalk and which is less one block from an intersection at which traffic control signals are in operation”.

a rule “No driver of a vehicle shall drive such vehicle on, over, or across any fire hose laid on any street or private road, unless directed so to do by the person in charge of such hose or a police officer”